Two-objective evaluation framework for single-cell foundation model interpretability

Objective A — GRN recovery

Edge-identification task

Do attention edges identify curated regulatory interactions?

Evaluation: AUROC vs. TRRUST/DoRothEA, top-K F1, CSSI

Objective B — Perturbation-target prediction

Effect-magnitude task

Do attention edges predict which genes show DE after a CRISPR perturbation?

Evaluation: per-perturbation AUROC, incremental value

Mathematically distinct: an edge can be a true direct regulator yet add little to Objective B (e.g. buffered), and a non-regulatory feature can score on Objective B if it proxies target susceptibility.

(i) Trivial baseline

variance, mean, dropout — no model

→ Objective B

(ii) Incremental value

edges + gene features, hard CV

→ Objective B

(iii) Residualisation & propensity match

remove gene-level confounds

→ Both objectives

(iv) Causal ablation

head mask, MLP zero, fidelity diag.

→ Objective B

(v) Cross-context

K562, RPE1, T cells, CRISPRi/CRISPRa

→ Both objectives

Positive control: pipeline sensitivity

Synthetic Perturb-seq with planted ground truth + real-data per-TF AUROC against TRRUST direct targets

Confirms: pipeline detects pairwise signal when it exists

What each family can conclude

(i)–(iv) → no incremental Obj-B value beyond gene features
(iii)+CSSI → Obj-A signal is recoverable, layer-specific

(v) → context-dependent on raw scores, but Obj-B null is universal

Convergent two-part conclusion

Cautionary (Objective B):
attention edges add no incremental value over gene-level features for predicting DE

Constructive (Objective A):
layer-specific regulatory structure exists; CSSI recovers it, up to 1.85× over baseline